

Kmos-0207

SUMMARY

The Governance Service is the authoritative trust engine of the Knowledge & Media Operating System (KMOS). Its responsibility is to establish, evaluate, record, and preserve institutional trust.

VERSION

1.0

STATUS

Draft

PURPOSE

The Governance Service SHALL govern: Policies; Approvals; Reviews; Certification; Compliance; Evidence Evaluation; Decision Records; Audit; Risk; Exceptions; Institutional Accountability. The Governance Service SHALL determine whether institutional work is sufficiently trustworthy to progress through its lifecycle.

MISSION

Provide explainable, evidence-based institutional trust. The Governance Service answers the question: "Can this organization trust this result, and why?"

RESPONSIBILITIES

The Governance Service SHALL own: Policies; Policy Versions; Approvals; Review Records; Certification Records; Compliance Records; Decision Records; Risk Assessments; Exception Records; Governance Lifecycle; Evidence Evaluation; Audit Decisions; Approval Workflows; Trust Assessments; Governance Metrics.

NON-RESPONSIBILITIES

The Governance Service SHALL NOT: Store Knowledge Objects; Store Assets; Execute Capabilities; Coordinate Workflows; Authenticate Identities; Publish Media; Maintain Search Indexes; Perform Business Processing. The Governance Service governs institutional decisions. Business execution remains elsewhere.

CANONICAL OBJECTS OWNED

Policy; Policy Version; Approval; Review; Certification; Compliance Record; Decision; Risk Assessment; Exception; Governance Audit; Trust Assessment. Future governance objects SHALL extend KMOS-0190.

ENGINEERING PHILOSOPHY

Governance SHALL be evidence-driven. Trust SHALL be explainable. Approvals SHALL be reproducible. Policies SHALL be versioned. Every governance decision SHALL preserve: Reason; Evidence; Reviewer; Authority; Policy Version; Time; Context. Institutional trust SHALL never rely upon undocumented judgment.

INTERNAL MODULES

Policy Registry; Approval Manager; Review Manager; Certification Manager; Compliance Manager; Decision Engine; Risk Manager; Exception Manager; Audit Manager; Evidence Evaluator; Governance API; Governance Event Publisher; Metrics Engine. Each module SHALL have one clearly defined responsibility.

PUBLIC BUSINESS APIS

Register Policy; Retrieve Policy; Evaluate Policy; Create Review; Complete Review; Approve Decision; Reject Decision; Create Certification; Revoke Certification; Record Compliance; Create Exception; Close Exception; Retrieve Audit Record; Retrieve Trust Assessment; Retrieve Governance Metrics; Export Governance Report. Business APIs SHALL expose governance concepts rather than implementation details.

CANONICAL EVENTS PUBLISHED

PolicyRegistered; PolicyUpdated; ApprovalRequested; ApprovalGranted; ApprovalRejected; ReviewCompleted; CertificationGranted; CertificationRevoked; ComplianceRecorded; RiskAssessed; ExceptionCreated; ExceptionResolved; TrustAssessmentCompleted; GovernanceAuditCreated; DecisionRecorded. Every governance transition SHALL publish canonical events.

CANONICAL EVENTS CONSUMED

KnowledgeUpdated; AssetRegistered; WorkflowCompleted; CapabilityRegistered; PublicationReleased; IdentityAuthenticated; ConfigurationChanged; EvidenceCollected. Governance SHALL evaluate business events rather than business logic.

POLICY REGISTRY

The Policy Registry SHALL maintain: Canonical Identifier; Business Purpose; Scope; Version; Owner; Lifecycle; Dependencies; Approval Requirements; Documentation. Policies SHALL remain independently versioned.

APPROVAL MANAGEMENT

Approvals SHALL support: Single Reviewer; Multiple Reviewers; Sequential Approval; Parallel Approval; Conditional Approval; Delegated Approval; Escalation. Approval SHALL preserve institutional accountability.

REVIEW MANAGEMENT

Reviews MAY include: Editorial Review; Technical Review; Knowledge Review; Legal Review; Compliance Review; Publication Review; Quality Assurance. Review SHALL preserve evidence supporting conclusions.

CERTIFICATION

Certification SHALL evaluate: Knowledge Quality; Capability Compliance; Workflow Conformance; Asset Integrity; Documentation; Testing; Operational Readiness. Governance SHALL preserve certification history.

COMPLIANCE

Compliance SHALL support: Institutional Policies; Industry Standards; Regulatory Requirements; Retention Policies; Privacy Requirements; Internal Procedures. Compliance SHALL remain policy-driven.

RISK ASSESSMENT

Every governance decision MAY include: Risk Level; Business Impact; Likelihood; Mitigation; Residual Risk; Approval Requirements. Risk SHALL remain queryable throughout the institutional record.

EXCEPTIONS

Exception records SHALL preserve: Reason; Approver; Scope; Duration; Compensating Controls; Evidence; Closure. Exceptions SHALL remain visible throughout their lifecycle.

TRUST ASSESSMENT

Trust SHALL be evaluated using institutional evidence, including: Knowledge Provenance; Asset Integrity; Workflow Completion; Capability Certification; Reviewer Approval; Policy Compliance; Identity Verification. Trust SHALL be explainable.

SECURITY

The Governance Service SHALL support: Authentication; Authorization; Policy Protection; Audit Logging; Decision Integrity; Approval Validation; Evidence Protection. Security SHALL conform to KMOS-0190.

OBSERVABILITY

The Governance Service SHALL publish: Approval Metrics; Policy Metrics; Certification Metrics; Compliance Statistics; Risk Statistics; Exception Metrics; Trust Assessment Metrics; Operational Health; Institutional Trust Metrics.

SCALABILITY

The Governance Service SHALL scale independently across: Policies; Approvals; Reviews; Organizations; Certifications; Compliance Records; Risk Assessments; Audit Volume. Scaling SHALL remain independent of business domains.

FAILURE STRATEGY

Failures SHALL never compromise institutional trust. Recovery SHALL support: Policy Reconstruction; Approval Recovery; Audit Preservation; Decision Verification; Operational Alerts; Evidence Validation. Trust SHALL remain recoverable.

TESTING STRATEGY

Testing SHALL include: Policy Evaluation; Approval Workflows; Certification; Compliance; Risk Assessment; Exception Handling; Audit; Replay; Performance; Migration; Governance Integrity.

ACCEPTANCE CRITERIA

A compliant Governance Service SHALL: Maintain policies; Support approvals; Maintain certifications; Maintain audit records; Support compliance; Support risk assessment; Support trust evaluation; Publish canonical events; Support KMOS-0190; Remain implementation independent.

REALITY CHECK

Governance reviews — Media Pipeline governance process — Proven; PASS / FAIL certification — Release gates — Proven; Evidence-based approval — Implementation verification — Proven; Capability certification — Existing governance — Partial; Policy registry — Planned — Early; Automated trust assessment — Planned — Not Implemented; Risk management — Governance discussions — Partial; Exception management — Planned — Not Implemented. The implementation SHALL prioritize proven governance workflows before introducing automated governance capabilities.

REFERENCE IMPLEMENTATION

governance-service/ with api/, application/, domain/(policies, approvals, reviews, certification, compliance, risk, trust, exceptions, audit), events/, repositories/, evaluators/, tests/, migrations/, documentation/. Governance SHALL remain independent of workflow implementation. Workflow coordinates governance. Governance evaluates trust.

CLAUDE IMPLEMENTATION GUIDE

Objective: Implement the authoritative Governance Service for KMOS. Primary Specifications: KMOS-0190, KMOS-0200, KMOS-0201, KMOS-0202, KMOS-0203, KMOS-0204, KMOS-0206. Deliverables: Production-ready Governance Service; Policy Registry; Approval Manager; Certification Manager; Risk Assessment Engine; Compliance Manager; Trust Assessment Engine; Automated Tests; Deployment Manifests; Operational Documentation. Acceptance: The implementation SHALL satisfy every requirement defined in this specification. Every governance decision SHALL be explainable through preserved evidence. Institutional trust SHALL emerge from documented facts rather than implicit assumptions.

RELATIONSHIP TO OTHER SERVICES

Knowledge Service (evaluates knowledge quality and approval); Asset Registry (evaluates evidence integrity and chain of custody); Event Service (consumes business events and publishes governance decisions); Workflow Service (coordinates governance activities and approval tasks); Capability Registry (certifies executable business capabilities); Identity Service (provides authenticated and authorized participants); Configuration Service (provides governance configuration and operational policies). The Governance Service SHALL remain the authoritative owner of institutional trust.

ARCHITECTURAL CONSEQUENCES

Trust becomes measurable. Approvals become reproducible. Compliance becomes observable. Policies become versioned. Certifications become auditable. Risk becomes explicit. Institutional decisions remain explainable for decades. The Governance Service becomes the institutional trust engine of KMOS.

REFERENCE EXAMPLE

A corrected transcript is submitted for publication → Workflow requests governance review → Knowledge provenance is evaluated → Supporting Assets are verified → Capability certification is confirmed → Reviewer identity is authenticated → Editorial approval is completed → Policy compliance is verified → Trust assessment succeeds → Publication is authorized → Governance decision is permanently recorded. Years later, the organization can reconstruct not only what was published, but precisely why it was considered trustworthy.

COMPLETION OF THE FOUNDATIONAL INSTITUTIONAL ENGINES

KMOS-0201 through KMOS-0207 define seven foundational engines: Meaning Engine (Knowledge Service); Evidence Engine (Asset Registry Service); Communication Engine (Event Service); Coordination Engine (Workflow Service); Execution Engine (Capability Registry Service); Accountability Engine (Identity Service); Trust Engine (Governance Service). These engines provide the permanent institutional foundation for domain services, AI capabilities, media processing, publishing systems, preservation workflows, learning platforms, research tools, and future KMOS applications, and are intended to remain stable across technology changes, implementation languages, deployment environments, and future generations of AI systems.

NEXT SPECIFICATION

KMOS-0208 defines the Search & Discovery Service, which SHALL provide semantic, evidence-aware, governance-aware, and capability-aware discovery across canonical objects managed by the platform, enabling humans, applications, workflows, and AI collaborators to discover institutional knowledge while respecting governance, security, provenance, and trust.